

Stability :-

$$A_f(s) = \frac{A(s)}{1 + \beta A(s)A(s)}$$

$$\approx \frac{A(s)}{1 + \beta A(s)}$$

$$A_f(j\omega) = \frac{A(j\omega)}{1 + \beta A(j\omega)}$$

$$A(s) = \frac{A_0}{(1 + \frac{s}{\omega_{p1}})(1 + \frac{s}{\omega_{p2}})} = \frac{A_0}{(1 + \frac{j\omega}{\omega_{p1}})^2 \tan^{-1}(\frac{\omega}{\omega_{p2}})}$$

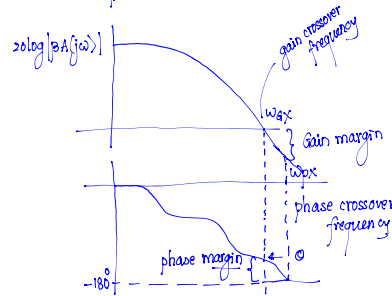
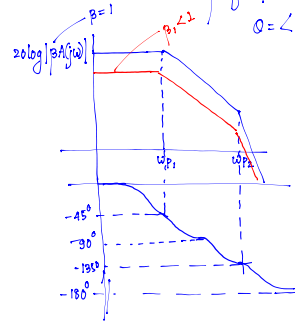
Loop gain

$$\beta A(s) = \beta A(j\omega)$$

$$= |\beta A(j\omega)| e^{j\theta}$$

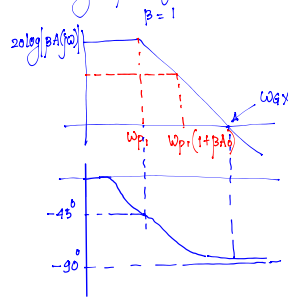
$$\theta = \angle \beta A(j\omega)$$

$$A(j\omega) = \frac{A_0}{\beta (1 + \frac{j\omega}{\omega_{p1}})}$$

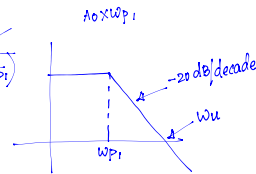


$$\text{phase margin (PM)} = 180^\circ + \angle \beta A(j\omega) \big|_{\omega=\omega_{gc}}$$

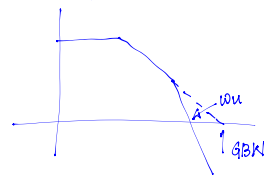
Single pole system :-



$$A(s) = \frac{A_0}{(1 + \frac{s}{\omega_{p1}})}$$



Two pole system



$$\text{PM} = 180^\circ + \angle \beta A(j\omega) \big|_{\omega=\omega_{gc}}$$

$$= 180^\circ - 90^\circ$$

$$= 90^\circ$$

$$A_f(j\omega) = \frac{A_0 / (1 + \frac{s}{\omega_{p1}})}{1 + \beta \frac{A_0}{1 + \frac{s}{\omega_{p1}}}}$$

$$= \frac{A_0}{1 + \frac{s}{\omega_{p1}} + \beta A_0}$$

$$= \frac{A_0}{(1 + \beta A_0) (1 + \frac{s}{\omega_{p1}(1 + \beta A_0)})}$$

Single pole amplifiers are always unconditionally stable

Two pole systems

